EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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Abstract

Clinical NLP pipelines detect assertion status—negation, uncertainty, family history—with $F_1 > 0.90$ [1], yet no existing system propagates these labels through knowledge graph construction into retrieval-augmented generation. We call this the *epistemic propagation gap*: when assertion labels are discarded during graph construction, downstream generation cannot faithfully represent the epistemic state of clinical findings. We present EpiKG, a clinical knowledge graph that closes this gap with a 7-value assertion schema on every edge, a tri-temporal model (valid time, transaction time, and NLP-asserted temporality), and an intent-aware retrieval layer that dispatches change, current-state, and historical questions to specialized graph-traversal strategies. On ClinicalBench ($n = 400$, MIMIC-IV, MedGemma 27B), EpiKG unlocks reasoning categories that are otherwise inaccessible: change-detection accuracy rises from 0% to 60% ($+60$ pp), family-history from 0% to 57% ($+57$ pp), and current-state from 64% to 76% ($+12$ pp). These gains are concentrated: in aggregate, intent-aware KG-RAG does not lift overall accuracy (-5.5 pp), because retrieval introduces noise in categories where the LLM is already strong. Notably, vanilla RAG *without* epistemic structure drops accuracy by 9.0 pp—assertion-blind retrieval is strictly worse than no retrieval. These results establish that the right evaluation lens for clinical KG-RAG is the category$\times$condition interaction, not the aggregate, and that epistemic structure is the difference between retrieval that helps and retrieval that harms.

1 Introduction

A physician documents "patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated"—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [2, 1], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because no system, to our knowledge, propagates assertion labels beyond the extraction stage into knowledge representation or retrieval. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing "patient has diabetes" from "rule out diabetes." Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen's interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.5, Appendix B), and realize the invariant with a seven-value schema extending i2b2 [2], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.

2. **Tri-temporal KG with Allen's interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine temporal relations (seven from Allen's algebra [12] plus CONCURRENT and UNKNOWN) as first-class attributes, enabling temporal reasoning over longitudinal records.

3. **Five-condition ablation with bookend baselines.** We evaluate five retrieval conditions—C1 (no retrieval) through C4g (intent-aware KG-RAG)—on ClinicalBench and SliceBench. Two bookend baselines, C6 (brute-force full-context) and C7 (structured-only, no LLM), are defined to bound the design space; their evaluation is planned as future work.

4. **ClinicalBench and SliceBench.** Two benchmarks (400 and 144 questions) across 9 epistemic categories target the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. A pilot physician adjudication ($n = 5$) reveals systematic over-strictness in automated scoring; a full two-physician study is planned.

The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* The answer is nuanced: KG-RAG unlocks reasoning categories that are otherwise inaccessible—change detection, family-history attribution, cross-admission comparison—while it can hurt on categories where the LLM's parametric knowledge already suffices and added context introduces noise. Vanilla RAG without epistemic structure performs *worse* than no retrieval at all, establishing that assertion preservation is not optional but load-bearing. The magnitude and even direction of the effect depend on model strength, producing a category×condition×model interaction that aggregate accuracy alone obscures (Section 5).

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 7, Appendix).

Medical RAG systems. Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [15] emulates physician reasoning; MedRAG [16] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on edges. AutoRD [19] applies GPT-4 to rare disease literature; RECAP-KG [20] handles messy GP notes; FHIR-Ontop-OMOP [21] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

Assertion detection. NegEx [22] introduced trigger-based negation detection; ConText [23] extended it with temporality and experiencer. The i2b2/VA shared task [2] formalized a six-class taxonomy; Gul et al. [1] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

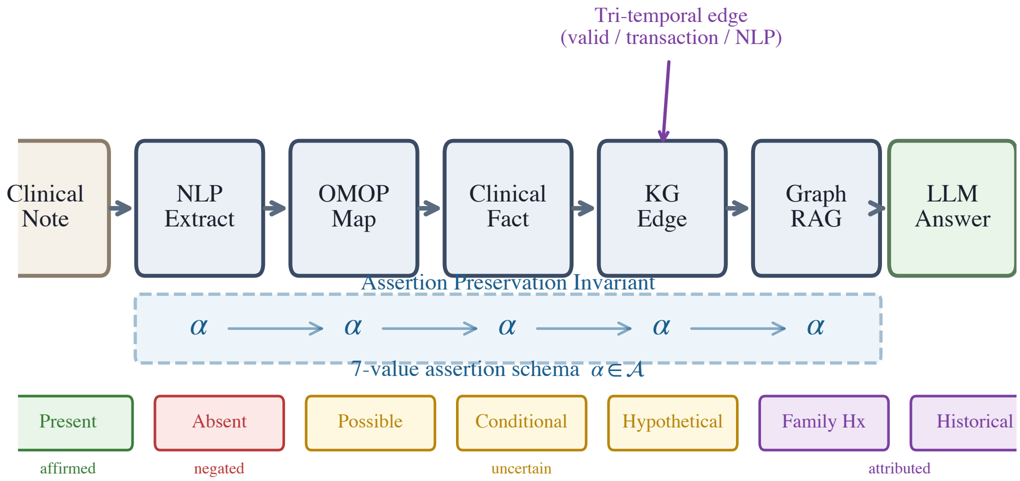

Figure 1: The EpiKG pipeline. The assertion label α (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [11] implements bitemporal edges [24] but lacks clinical ontology alignment; TEO [13] and TCL [14] formalize Allen's relations [12] as annotation ontologies or modal operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [25] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges, enabling cohort queries via edge scans.

3.2 Epistemic Assertion Schema

Clinical notes contain qualified statements—"*no evidence of pneumonia*," "*possible CHF*," "*mother had breast cancer*"—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR restricts assertion metadata to conditions alone. EpiKG defines a 7-value assertion taxonomy:

$$\alpha \in \{\text{Pres., Abs., Poss., Cond., Hypo., Fam.Hx., Hist.}\} \tag{1}$$

This extends the i2b2 six-class taxonomy [2] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates (definitions in Appendix O). A rule-based classifier (101 scope-aware trigger patterns) assigns α with an associated confidence score; we chose rule-based classification for deterministic behavior and interpretability, though fine-tuned LLM classifiers [1] could be substituted. Intrinsic classifier evaluation is deferred; the end-to-end ClinicalBench ablation (C3→C4) serves as a functional test of assertion propagation. The label is carried through every stage: from `ExtractedMention` through ensemble merging, `ClinicalFact` records, KG edge properties, and retrieval scoring. To our knowledge, no prior system [1, 26] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. Bi-temporal databases [24] distinguish valid time from transaction time; EpiKG adds a third dimension (NLP-asserted temporality) and stores all three on every KG edge:

Definition 1 (Tri-Temporal Edge)**.** *An edge* $e = (s, p, o, \alpha, \boldsymbol{\tau}_v, \boldsymbol{\tau}_t, \tau_a, r, c)$ *where:*

- s, o *are source and target nodes;* p *is the predicate (one of 24 edge types);*
- $\alpha \in \mathcal{A}$ *is the assertion label (Eq. 1);*
- $\boldsymbol{\tau}_v = (\textit{event\_date}, \textit{valid\_from}, \textit{valid\_to})$ *is the* ***valid time*** *interval—when the relationship held in the real world;*
- $\boldsymbol{\tau}_t = (\textit{recorded\_at}, \textit{doc\_date}, \textit{created\_at})$ *is the* ***transaction time****—when it was recorded;*
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ *is the* ***NLP-asserted temporality****;*
- $r \in \mathcal{R}$ *is an Allen interval algebra relation [12];* $c \in [0, 1]$ *is temporal confidence.*

$\mathcal{R}$ comprises seven of Allen's 13 canonical relations [12] (merging symmetric pairs like Before/Meets) plus CONCURRENT and UNKNOWN (nine total; full mapping in Appendix P). Unlike TEO [13] and TCL [14], which use Allen's relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal.

3.4 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π, the retrieval pipeline: (1) extracts concepts from q via NLP + OMOP enrichment; (2) traverses 2–3 hops via bidirectional BFS over patient KG edges and OMOP vocabulary relationships (20M+ edges) via PostgreSQL CTEs (Appendix J), pruning edges with $c < 0.3$; (3) groups edges into four temporal views (event timeline, current state, historical, conflicts); (4) retrieves matching guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{1}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{1}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \tag{2}$$

where the bonuses for question-relevant edge types (0.2) and current temporality (0.1) were set by manual tuning on a development set of 20 questions. The surviving subgraph is serialized into structured text preserving assertion labels (e.g., "ABSENT: pneumonia"); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

3.5 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [27]. The *epistemic state* of a clinical mention m is the tuple $e(m) = (c, \alpha, \xi, \tau)$: OMOP concept, 7-value assertion label, experiencer ($\{\text{PATIENT}, \text{FAMILY}\}$), and temporality ($\{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$). A pipeline *epistemically preserves* m if the output epistemic state equals the extraction state at every stage. We formalize (Appendix B) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss $\Delta H = H(A_c^\pi) \geq 0$ [28], and that assertion-faithful accuracy is bounded by $1 - f_{np}(c)$, where f_{np} is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

3.6 Intent-Aware Retrieval (C4g)

The multi-hop BFS described above treats all questions uniformly, yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

Table 1: ClinicalBench ablation conditions. C1–C4g form the primary ablation ladder.

ID	Condition	Retrieval	Assertion	Temporal
C1	LLM Alone	None	None	None
C2	+ Vanilla RAG	Document	None	None
C3	+ KG-RAG	Graph + Doc	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal

Intent classifier. A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Algorithm 1, Appendix Q).

Change retrieval. For $\iota = $ CHANGE, edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs (k, k')—concepts *added* ($\mathcal{C}_{k'} \backslash \mathcal{C}_k$), *removed* ($\mathcal{C}_k \backslash \mathcal{C}_{k'}$), and *continued* ($\mathcal{C}_k \cap \mathcal{C}_{k'}$)—and returns structured evidence.

Current-state and historical retrieval. For CURRENT_STATE, the module filters to edges with $\tau_a = $ CURRENT or open-ended validity, deduplicates by concept, and emits "NOT FOUND" for missing concepts. For HISTORICAL, it selects edges with $\tau_a = $ PAST and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled "resolved"). Each intent triggers a type-specific prompt template that structures evidence according to question semantics (Appendix S).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [29]) or agent task completion [30] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [31].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*. Gold-standard answers were created through manual chart review and verified against source documents; a pilot physician adjudication ($n = 5$, Appendix M) assessed quality. Five ablation conditions progressively add system components (Table 1), plus a diagnostic condition C5 (Appendix R). Two bookend baselines—C6 (brute-force full-context, infeasible for MedGemma's 8K context window on complex patients) and C7 (deterministic KG only)—bracket the design space; their evaluation with long-context models is planned as future work.

Endpoints. The **pre-registered primary endpoint** is the hard longitudinal subset (*change* + *current_state* + *historical*, $n = 130$): C4g vs. C1 aggregate accuracy. The **secondary endpoint** is full 400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas and the C4→C4g transition on the hard subset.

What each transition isolates. C3→C4 isolates the *assertion preservation effect*: both use graph retrieval, but only C4 carries the 7-value assertion schema and tri-temporal model. C4→C4g isolates the *intent-aware routing effect*: both carry full assertion metadata, but C4g dispatches question-type-specific graph traversals.

4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24 questions (including hard longitudinal categories: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a monotone context progression from parametric-only to full system (Appendix I). The critical comparison

Table 2: ClinicalBench ablation results (400 questions, MedGemma 27B, deterministic keyword evaluator). Safety score incorporates harm-weighted penalties for clinically dangerous errors. Best per column in **bold**.

	Condition	Accuracy	Safety
C1	LLM Alone	**71.5%**	0.752
C2	+ Vanilla RAG	62.5%	0.643
C3	+ KG-RAG (no assertion)	59.2%	—†
C4	+ Epistemic KG-RAG	**71.5%**	**0.775**
C4g	+ Intent-Aware KG-RAG	66.0%	0.643
C5	+ Full System	68.2%	—†

†Safety scores for C3 and C5 were not computed in the final evaluator run.

is B2→B3: both have the same documents, but B3 adds structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

ClinicalBench uses a *deterministic keyword evaluator* that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers (keyword lists in Appendix N). This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench results. The answering model is MedGemma 27B (primary) or Claude Opus 4.6 (cross-model validation).

SliceBench uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [32]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness.

Statistical reporting. We report BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) on SliceBench aggregate metrics (Table 4); CIs are computed at the question level. ClinicalBench results are reported as point estimates with within-run paired comparisons; CIs are omitted because the primary analysis targets condition deltas rather than absolute levels. A safety score with asymmetric weighting ($w = 2.0$ for false-positive assertion errors) captures clinical risk (Appendix D).

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers. ClinicalBench uses a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3). SliceBench reports BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) [33], resampled at the question level. All ClinicalBench results are *within-run paired comparisons*: conditions were evaluated in the same run with the same evaluator version, ensuring deltas are internally valid even when absolute levels vary across runs (Appendix N).

5.1 ClinicalBench: Ablation Results

Adding vanilla RAG *hurts*: C2 underperforms C1 by 9.0 pp (-0.109 safety score). Assertion-blind KG-RAG (C3) scores lowest (59.2%)—structured paths presenting negated conditions as graph edges actively mislead the model. The C3→C4 transition (+12.3 pp) isolates the assertion preservation effect: both conditions use graph retrieval, but only C4 carries the 7-value assertion schema, recovering to 71.5% with the highest safety score (0.775)—matching C1's accuracy while producing fewer clinically dangerous errors. Intent-aware KG-RAG (C4g) scores 66.0% overall—5.5 pp below C1—but this aggregate masks its transformative effect on longitudinal categories.

5.2 Category×Condition Interaction

Table 3 and Figure 2 reveal a *category×condition interaction*: structured epistemic context is transformative for questions requiring cross-admission normalization but can hurt where the LLM's parametric knowledge already suffices (per-category delta chart in Figure 4, Appendix; full results including C5 in Table 13, Appendix).

Table 3: ClinicalBench per-category accuracy (%) across conditions (MedGemma 27B). Longitudinal categories (below midrule) show the strongest category×condition interaction. Best per row in **bold**.

Category	n	C1	C2	C3	C4	C4g	C1→C4g
Negation	110	**100.0**	93.6	81.8	**100.0**	86.4	↓ 13.6 pp
Conditional	20	**100.0**	90.0	55.0	70.0	65.0	↓ 35.0 pp
Uncertainty	40	37.5	2.5	7.5	**57.5**	32.5	↓ 5.0 pp
Family history	30	0.0	0.0	53.3	**70.0**	56.7	↑ 56.7 pp
Sequence	40	80.0	80.0	80.0	87.5	**90.0**	↑ 10.0 pp
Current state	50	64.0	54.0	50.0	54.0	**76.0**	↑ 12.0 pp
Duration	30	**100.0**	**100.0**	83.3	83.3	46.7	↓ 53.3 pp
Historical	50	**94.0**	78.0	68.0	60.0	40.0	↓ 54.0 pp
Change	30	0.0	0.0	3.3	3.3	**60.0**	↑ 60.0 pp
Overall	**400**	**71.5**	62.5	59.2	**71.5**	66.0	↓ 5.5 pp

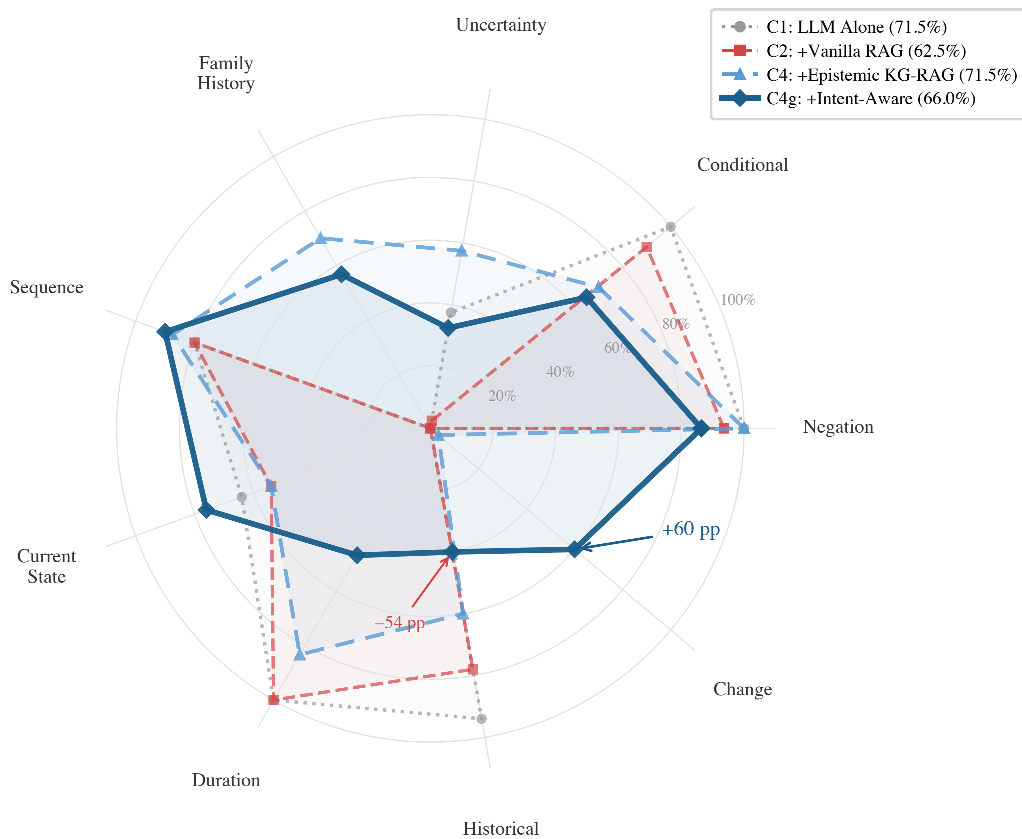

Figure 2: ClinicalBench per-category accuracy across four conditions (all 9 categories). C4g (dark solid) uniquely unlocks change detection (+60 pp) but loses on historical (−54 pp) and duration—the category×condition interaction that aggregate accuracy obscures.

Where intent-aware routing excels. The "change" category—near zero for *all* conditions C1 through C5—jumps to 60.0% under C4g (+60.0 pp vs. C1). Change questions require comparing a clinical concept across admissions, which is impossible from a single note (C1), missed by chunk-level retrieval (C2), and lost in assertion-unaware graph paths (C3). Even C4's epistemic metadata is insufficient; C4g's intent classifier routes to admission-partitioned retrieval. Current state benefits (+12.0 pp), family history rises from 0% to 57%, and sequence gains +10.0 pp.

Where intent-aware routing hurts. C4g loses on historical (−54.0 pp), duration (−53.3 pp), and conditional (−35.0 pp)—categories where C1 was at or near ceiling and injected context overrides

Table 4: SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

Table 5: ClinicalBench cross-model aggregate results (400 questions, keyword evaluator).

	Model	C1	C4	Δ
Open	MedGemma 27B	71.5%	71.5%	+0.0 pp
Commercial	Claude Opus 4.6	72.5%	70.2%	−2.2 pp

correct reasoning [34]. The pre-registered primary endpoint was not met: on the hard longitudinal subset ($n = 130$), C4g (58.5%) does not exceed C1 (60.8%). However, the diagnostic C4→C4g transition (+13.8 pp on this subset) isolates intent-aware routing as the mechanism enabling category-specific longitudinal gains, even as historical regression (−20.0 pp) offsets the change improvement (+56.7 pp).

5.3 SliceBench and Cross-Model Validation

Document retrieval provides a large benefit (B0→B2 = +34.8 pp). The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance. However, tier-stratified point estimates show a complexity-dependent trend: +0.6 pp (Tier A, 1–2 encounters), +1.0 pp (Tier B, 5–10), +5.0 pp (Tier C, 15+), consistent with the hypothesis that KG context is most valuable when document volume overwhelms chunk-level retrieval (Appendix K, Figure 3).

Cross-model validation. We ran ClinicalBench with Claude Opus 4.6 alongside MedGemma 27B (Table 5). Under C4, MedGemma shows no aggregate change (+0.0 pp) and Opus shows a small regression (−2.2 pp). Both models show the same directional pattern: gains on assertion-dependent categories (uncertainty, family history for MedGemma; negation, duration for Opus) and losses where parametric knowledge suffices (Appendix Table 8). The "change" category remains near zero for C4 on both models (3.3%, 13.3%), motivating C4g's intent-aware routing.

An experiencer attribute propagation ablation (Appendix F) confirmed that the `experiencer` field is load-bearing: propagating it through multi-hop traversal improved family history by +10.0 pp with zero regressions, halving the C4-vs-C1 gap from −4.2 pp to −2.2 pp.

6 Discussion and Limitations

Why vanilla RAG hurts. Clinical notes contain dense negated, uncertain, and historical findings. When vanilla RAG retrieves "patient denies diabetes," the word "diabetes" enters context without the epistemic frame distinguishing absence from presence, actively misleading the model. The C2 $<$ C1 finding (−9.0 pp, Table 2) is consistent with GraphRAG-Bench [34], which reports that GraphRAG underperforms when graph structure fails to encode task-relevant semantics—here, assertion status.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. For simple patients with few notes, the LLM tracks assertion context within retrieved chunks and the KG adds little. C4g's intent-aware routing provides direct evidence for this mechanism: the "change" category—which was near-zero across all prior conditions including C4—jumps to 60% under C4g (+60 pp vs. C1), confirming that *targeted graph traversal strategy*, not merely graph structure, is the key factor. By routing change questions to cross-admission edge partitioning and current-state questions to recency-weighted retrieval, C4g demonstrates that matching retrieval mode to question intent unlocks gains that uniform retrieval cannot.

Model-strength interaction. The cross-model results (Table 5) show that epistemic KG-RAG's benefit is inversely related to baseline model capability. Under the uniform C4 condition, MedGemma 27B showed no aggregate change ($+0$ pp) and Opus showed a small regression (-2.2 pp). However, C4g reveals a *category-specific* interaction: under intent-aware routing, MedGemma gains $+60$ pp on change and $+57$ pp on family history while losing ground where its C1 baseline is already adequate. Under C4, Opus gains on assertion-dependent categories (negation $+2.7$ pp, duration $+10.0$ pp, uncertainty $+7.5$ pp) but regresses on conditional (-15.0 pp) and family history (-10.0 pp). This reveals a design trade-off: structured context is transformative for cross-admission reasoning but can override correct answers where parametric knowledge suffices. Model-adaptive evidence presentation (e.g., confidence-gated injection) is a promising direction.

Limitations. Sample size is modest: SliceBench covers 6 patients (144 questions) and ClinicalBench covers 45 patients (400 questions), both from a single site (MIMIC-IV [31]). The pre-registered primary endpoint (C4g vs. C1 on the hard longitudinal subset) was not met; category-specific gains are offset by regressions where parametric knowledge suffices. The overall B2→B3 CI crosses zero ([-1.5 pp, +5.9 pp]); tier-level estimates ($n = 2$ patients per tier) are hypothesis-generating only. Bootstrap CIs are at the question level; patient-level clustering would be more conservative and is planned. The evaluation is an ablation study; head-to-head comparisons with existing medical RAG systems [5, 6, 15] were not performed because these target population-level or factoid QA rather than patient-level longitudinal reasoning, and lack open-source MIMIC-IV implementations. Cross-model validation covers C4 only, not C4g; intent-aware routing generalization remains an open question. C5 showed degradation (68.2% vs. C4's 71.5%), likely from non-relevant module activation. SliceBench's LLM-as-judge evaluation may introduce biases; a pilot human validation ($n = 5$, single reviewer; Appendix M) revealed overly strict automated scoring on uncertainty and sequence categories. ClinicalBench uses a deterministic keyword evaluator to avoid this, but keyword matching may miss semantically correct answers that use different phrasing. A full two-physician blinded adjudication covering both benchmarks is planned but not yet complete. Run-to-run variance in MedGemma results (± 10 pp) arises from GPU non-determinism in Ollama's quantized inference (4-bit GGUF); all reported deltas rely on within-run paired comparisons to control for this. C4g's aggregate accuracy (-5.5 pp vs. C1) illustrates a key tension: structured context can *hurt* in the aggregate even while providing transformative category-level gains ($+60$ pp on change, $+57$ pp on family history), because formatting overhead degrades categories where the LLM already performs well. More broadly, systems like EpiKG could improve clinical decision support by surfacing cross-admission patterns that clinicians may miss under time pressure; however, over-reliance on automated KG-driven summaries risks anchoring bias if the underlying NLP extraction or assertion classification introduces systematic errors, and evaluation on a single US academic medical center (MIMIC-IV) leaves generalization to other settings unvalidated (Appendix T).

Future work. (1) Model-adaptive evidence presentation to reduce regressions on strong-baseline categories. (2) Improving evidence formatting (the 60% change-detection ceiling and historical's 94%→40% drop suggest formatting, not retrieval, as the bottleneck). (3) Expanding Tier C to 20+ patients and running two-physician blinded adjudication.

Conclusion. The epistemic propagation gap—the systematic loss of assertion, temporality, and experiencer metadata between NLP extraction and downstream reasoning—is a real and measurable phenomenon: the information-theoretic bound (Proposition 3) shows that collapsing non-present assertions to present destroys recoverable clinical information, and our ClinicalBench results confirm the empirical consequence—assertion-blind retrieval (C2) degrades accuracy by 9.0 pp while assertion-aware retrieval (C4) preserves it. EpiKG addresses this gap through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal strategy to question semantics. The benefit is category-specific rather than uniform: change detection rises from 0% to 60% and family history from 0% to 57%, while categories where the LLM is already strong can regress. This category×condition interaction—not aggregate accuracy—is the right lens for clinical KG-RAG evaluation.

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A Notation Summary

Symbol	Meaning
$\alpha \in \mathcal{A}$	Assertion label (7 values, Eq. 1)
$\boldsymbol{\tau}_v$	Valid time (event date, valid from/to)
$\boldsymbol{\tau}_t$	Transaction time (recorded at, doc date, created at)
τ_a	NLP-asserted temporality (Current / Past / Future)
$r \in \mathcal{R}$	Temporal relation (9 values, Table 15)
c	Confidence score $\in [0, 1]$
ξ	Experiencer (Patient / Family)
$e(m)$	Epistemic state tuple (c, α, ξ, τ)
q, π	Clinical question, patient
ι	Question intent (Change / Current_State / Historical / Default)
$f_{np}(c)$	Fraction of non-present assertions for concept c

Table 6: Key notation used throughout the paper.

B Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle that aboutness—the relationship between information artifacts and the entities they represent—must be preserved across transformations [27].

Definition 2 (Epistemic State). *For a clinical mention m, the* epistemic state *is the tuple $e(m) = (c, \alpha, \xi, \tau)$, where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in \{\text{PATIENT}, \text{FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the temporality. A pipeline P* epistemically preserves *mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the output of every pipeline stage is identical to the state at extraction.*

Proposition 3 (Assertion Entropy Loss). *For concept c in patient π's record, let A_c^π denote the assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes. The assertion entropy is*

$$H(A_c^\pi) = -\sum_i p(\alpha_i) \log p(\alpha_i) \ . \tag{3}$$

An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribution with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [28], with $\Delta H > 0$ strictly whenever any mention carries a non-present assertion.

Proof. Under the assertion-blind mapping $\phi : \alpha_i \mapsto \text{PRESENT}$ for all i, the output distribution assigns probability 1 to PRESENT and 0 to all other classes, so $H(\phi(A_c^\pi)) = 0$. By non-negativity of Shannon entropy, $\Delta H = H(A_c^\pi) - 0 = H(A_c^\pi) \geq 0$, with equality iff all mentions already carry $\alpha = \text{PRESENT}$. $\square$

Corollary 4 (Faithfulness Bound). *Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records where negated and uncertain mentions are prevalent—e.g., "no pneumonia," "possible CHF"—this bound can be substantially below 1.*

Proof. Without assertion labels, any predictor must assign a single assertion class to all mentions of c. Choosing PRESENT (the majority class in clinical text) yields accuracy $1 - f_{np}(c)$; the $f_{np}(c)$ non-present mentions are necessarily misclassified. $\square$

C Gap Analysis

Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but

Table 7: Capability comparison across representative systems. ✓ = supported, ○ = partial, × = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert.-aware RAG	Experiencer	Allen's algebra	Multi-hop (≥3)
DoctorRAG [15]	×	×	×	×	×	×	×	×
GFM-RAG [5]	○	×	×	×	×	×	×	✓
MedRAG [16]	○	×	×	×	×	×	×	○
KARE [6]	○	×	×	×	○	×	×	✓
Multi-LLM KG [18]	×	✓	○	×	×	×	×	×
MedTKG [10]	✓	×	×	○	×	×	×	✓
Graphiti [11]	×	×	×	○	×	×	×	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	○

without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models. EpiKG's partial mark on multi-hop traversal ($\geq$ 3 hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Appendix J).

D Safety Score

The safety score $S = 1 - \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for future work.

E Cross-Model Per-Category Results

Table 8: ClinicalBench per-category accuracy (%) by model under C4 (epistemic KG-RAG). Epistemic evidence helps both models on assertion-dependent categories but hurts where the baseline is already strong.

		MedGemma 27B			Claude Opus 4.6		
Category	n	C1	C4	Δ	C1	C4	Δ
Negation	110	100.0	100.0	=	96.4	99.1	↑2.7 pp
Conditional	20	100.0	70.0	↓30.0 pp	80.0	65.0	↓15.0 pp
Uncertainty	40	37.5	57.5	↑20.0 pp	47.5	55.0	↑7.5 pp
Family hist.	30	0.0	70.0	↑70.0 pp	83.3	73.3	↓10.0 pp
Sequence	40	80.0	87.5	↑7.5 pp	97.5	95.0	↓2.5 pp
Current st.	50	64.0	54.0	↓10.0 pp	58.0	48.0	↓10.0 pp
Duration	30	100.0	83.3	↓16.7 pp	90.0	100.0	↑10.0 pp
Historical	50	94.0	60.0	↓34.0 pp	44.0	38.0	↓6.0 pp
Change	30	0.0	3.3	↑3.3 pp	23.3	13.3	↓10.0 pp
Overall	**400**	**71.5**	**71.5**	=	**72.5**	**70.2**	↓**2.2** pp

F Experiencer Attribute Propagation Ablation

The `experiencer` attribute distinguishes patient conditions from family member conditions; without it, the graph conflates the two, causing family history misattribution.

The fix improved family history by +10.0 pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experiencer attribute is load-bearing for assertion-sensitive reasoning.

Table 9: Impact of experiencer attribute propagation on Opus C4 (400 questions, same C1 baseline of 72.5%). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	**73.3%**	+10.0 pp
Uncertainty	50.0%	**55.0%**	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	**68.2%**	**70.2%**	+2.0 pp
C4 vs C1 gap	−4.2 pp	−2.2 pp	gap halved

G SliceBench Figures

H Per-Category Delta Chart

I SliceBench Conditions

Table 10: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

J System Implementation Details

LLM baseline. On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [30]), establishing that the Claude model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench cross-model, SliceBench judge)—provides a reasonable LLM baseline (Table 11).

Model	**Accuracy**	**Step 1**	**N**
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 11: MedQA-USMLE results. Our LLM-alone baseline is competitive.

KG scale and traversal latency. The deployed system processes 145 documents across 85 patients, materializing 3,100 KG nodes and 8,803 edges. Graph traversal latency is sub-millisecond at this scale: 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 clinical calculators, 1,202 guideline sections, 20M+ OMOP vocabulary relationships, 101 assertion trigger patterns, and 24 edge types across 13 node types.

Multi-hop traversal. On the DR.KNOWS diagnostic reasoning benchmark [35], which measures KG traversal accuracy using PostgreSQL-backed clinical data, EpiKG achieves 0.420 overall (50% at 1-hop, 25% at 2-hop, 0% at 3-hop). Multi-hop degradation reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but scales poorly beyond 2 hops.

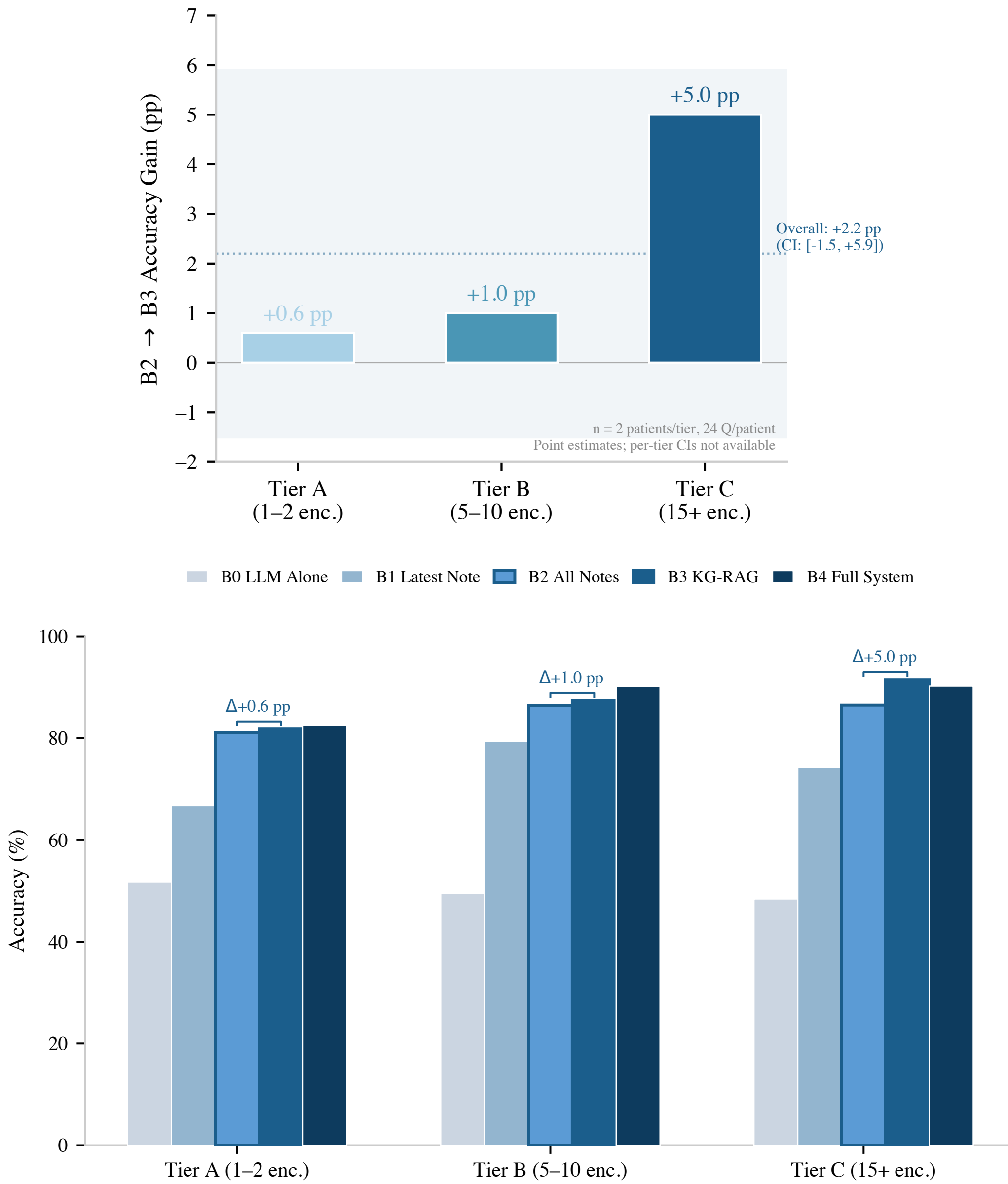

Figure 3: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer's incremental benefit (B2→B3) shows a directional, complexity-dependent trend: $+0.6$ pp (Tier A) to $+5.0$ pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is $+2.2$ pp (CI: [-1.5 pp, +5.9 pp]).

K SliceBench Tier-Stratified Results

L ClinicalBench Full Per-Category Results

M Human Validation Pilot

To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-certified physician who scored system outputs on a 4-point scale without knowledge of condition labels. The pilot ($n = 5$) revealed overly strict automated scoring on uncertainty and sequence questions. Gold standards were fully correct in 3/5 cases; model answers were partially correct or better in 3/5 cases where the auto-scorer marked them incorrect. A full two-physician blinded adjudication ($n \geq 30$) is planned but not yet complete.

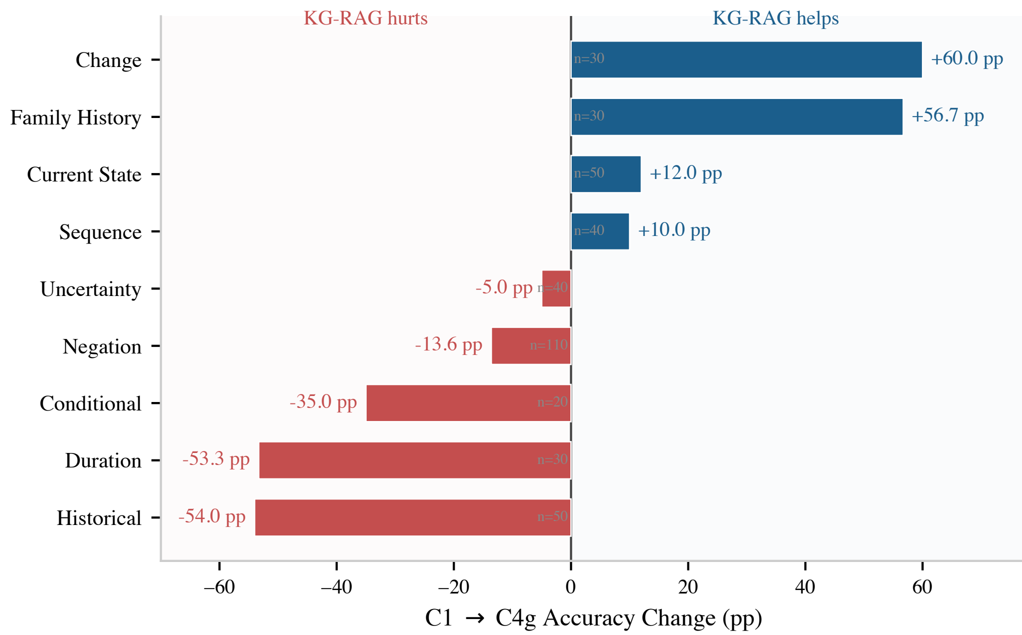

Figure 4: C1→C4g per-category accuracy change. Intent-aware KG-RAG produces large gains on categories requiring cross-admission normalization (change, family history) and losses on categories where the LLM's parametric knowledge already suffices (historical, duration, conditional). Sample sizes per category shown.

Table 12: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution.

		Score (%)					
Tier	Encounters	B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

Category	**C1**	**C2**	**C3**	**C4**	**C4g**	**C5**
Negation	100.0	93.6	81.8	100.0	86.4	98.2
Conditional	100.0	90.0	55.0	70.0	65.0	75.0
Uncertainty	37.5	2.5	7.5	57.5	32.5	40.0
Family History	0.0	0.0	53.3	70.0	56.7	56.7
Sequence	80.0	80.0	80.0	87.5	90.0	77.5
Current State	64.0	54.0	50.0	54.0	76.0	58.0
Duration	100.0	100.0	83.3	83.3	46.7	73.3
Historical	94.0	78.0	68.0	60.0	40.0	68.0
Change	0.0	0.0	3.3	3.3	60.0	3.3
Overall	**71.5**	**62.5**	**59.2**	**71.5**	**66.0**	**68.2**

Table 13: Complete ClinicalBench per-category accuracy (%, deterministic keyword evaluator) for all six conditions.

N Reproducibility Details

Models. ClinicalBench main ablation: MedGemma 27B (`gemma3:27b`, 4-bit GGUF quantization) via Ollama with deterministic keyword evaluator. Cross-model: MedGemma 27B and Claude Opus 4.6 (claude-opus-4-20250514) with the same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929, answering), Claude Opus 4.6 (judging). Temperature 0 throughout; 4-bit quantization introduces GPU-level non-determinism (± 10 pp run-to-run variance), controlled via within-run paired comparisons.

Evaluation. The deterministic keyword evaluator performs exact word-boundary matching (
`bKEYWORD`
`b`) against gold-standard assertion and temporal keywords. Per-category keyword sets: negation ("no," "denies," "absent," "negative"), uncertainty ("possible," "suspected," "may," "likely," "consider"), temporal ("before," "after," "during," "changed," "new"), plus category-specific medical terms. Evidence preambles (echoed graph context) are stripped before matching to prevent false positives.

Retrieval. C2's vanilla RAG uses TF-IDF similarity over chunked clinical notes (512-token chunks, 64-token overlap), retrieving the top-5 chunks by cosine similarity. ClinicalBench conditions C1–C4g correspond to SliceBench conditions: C1≈B0, C2≈B2, C4≈B3, C5≈B4.

Bootstrap. $n = 2{,}000$ resamples, seed 42, BCa method [33].

Data. De-identified MIMIC-IV clinical notes under PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Compute. ClinicalBench: ∼3.6h (single GPU); Opus cross-model: ∼20min (API); SliceBench: ∼2h (API).

Evaluator evolution. Key refinements: (i) substring to word-boundary matching, (ii) evidence preamble stripping. All main-text numbers use the final version.

O Assertion Category Definitions

Assertion	Definition	Example
PRESENT	Condition currently affirmed	"has diabetes"
ABSENT	Condition explicitly negated	"denies chest pain"
POSSIBLE	Condition suspected, not confirmed	"possible pneumonia"
CONDITIONAL	Contingent on specific circumstances	"if febrile, start antibiotics"
HYPOTHETICAL	Discussed as a scenario	"would need dialysis if..."
FAMILY_HISTORY	Attributed to a family member	"mother had breast cancer"
HISTORICAL	Previously true, not necessarily current	"former smoker"

Table 14: The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY_HISTORY.

P Temporal Relation Mapping

The nine temporal relations $\mathcal{R}$ used on KG edges are derived from Allen's 13 canonical interval relations [12] by merging symmetric pairs.

$\mathcal{R}$ value	Allen source(s)	Meaning
BEFORE	Before, Meets	A ends before B starts
AFTER	After, Met-by	A starts after B ends
DURING	During	A entirely within B
CONTAINS	Contains	A entirely contains B
OVERLAPS	Overlaps, Overlapped-by	A and B partially overlap
STARTS	Starts, Started-by	A and B share start time
FINISHES	Finishes, Finished-by	A and B share end time
CONCURRENT	Equals	A and B have same interval
UNKNOWN	—	Relation undetermined

Table 15: Mapping from Allen's 13 interval relations to the 9 temporal relation values stored on KG edges.

Q Intent-Aware Routing Algorithm

The C4g classifier is rule-based, mapping questions to one of four intents using keyword patterns (e.g., "changed", "new since" trigger CHANGE; "currently", "active problem" trigger CURRENT_STATE). Algorithm 1 details the full retrieval procedure.

Algorithm 1: Intent-Aware Retrieval (C4g)

Input: Question q, patient π
Output: Structured evidence E

$\mathcal{C} \leftarrow \textsc{ExtractConcepts}(q)$; `// NLP + OMOP enrichment`
$\iota \leftarrow \textsc{ClassifyIntent}(q)$; `//` $\in \{\textsc{Change}, \textsc{CurrSt}, \textsc{Hist}, \textsc{Default}\}$
if $\iota = \textsc{Change}$ **then**
  Partition edges by admission: $\mathcal{E}_k \leftarrow \{e \mid \texttt{hadm\_id}(e) = k\}$;
  foreach *admission pair* (k, k') *with* $k < k'$ **do**
    $\mathcal{A} \leftarrow \mathcal{C}_{k'} \setminus \mathcal{C}_k$; $\mathcal{R} \leftarrow \mathcal{C}_k \setminus \mathcal{C}_{k'}$; $\mathcal{S} \leftarrow \mathcal{C}_k \cap \mathcal{C}_{k'}$;
  $E_g \leftarrow \textsc{FormatChange}(\mathcal{A}, \mathcal{R}, \mathcal{S})$;
else if $\iota = \textsc{CurrSt}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Current} \vee \text{open validity})$;
  Deduplicate by concept; emit "NOT FOUND" for missing $c \in \mathcal{C}$;
else if $\iota = \textsc{Hist}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Past})$;
  Augment: concepts in earlier admissions but absent from latest $\rightarrow$ "resolved";
else
  $E_g \leftarrow \textsc{BidirectionalBFS}(\pi, \mathcal{C},\ \text{hops} = 2\text{–}3,\ c_{\min} = 0.3)$;
$E_d \leftarrow \textsc{RetrieveDocuments}(\pi, \mathcal{C})$;
return $\textsc{Compose}(E_g, E_d,\ \text{template}(\iota))$;

R C5 Full System Methodology

C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., CHA_2DS_2-VASc, MELD). C5 achieves 68.2% overall, below C1 and C4 (71.5% each), likely because non-relevant components dilute the context window with information unrelated to the question.

S Illustrative Retrieval Example

To illustrate why intent-aware routing matters, consider a change question: *"What medications changed between this patient's first and second admissions?"*

C1 (LLM alone). The model sees only the current note, which mentions metoprolol, atorvastatin, and "discontinued lisinopril." Without prior admission records, it cannot determine what was *added* vs. *continued*, and may hallucinate prior medication lists.

C4g (intent-aware KG-RAG). The intent classifier routes to CHANGE retrieval, which partitions KG edges by `hadm_id` and computes set differences:

- **Added** (admission 2 only): atorvastatin 40mg [PRESENT]
- **Removed** (admission 1 only): lisinopril 10mg [PRESENT→ABSENT]
- **Continued**: metoprolol 25mg [PRESENT, both admissions]

The assertion labels (PRESENT, ABSENT) disambiguate: "discontinued lisinopril" is not a current medication but a historical one whose status changed—exactly the distinction the epistemic schema preserves.

T Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [31] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous clinical decision-making.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: **Yes.**

Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.3. SliceBench claims include bootstrap CIs; ClinicalBench claims use within-run paired comparisons (Section 4.3). Claims are scoped with "to our knowledge" qualifiers where appropriate.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **Yes.**

   Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, category-level trade-offs (C4 vs C1), and the "change" category capability gap.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: **Yes.**

   Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in Appendix B state all assumptions explicitly. These are information-theoretic characterizations with all steps included.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: **Yes.**

   Justification: Appendix N reports exact model versions (MedGemma 27B, Claude Sonnet 4.5, Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2{,}000$, seed 42, BCa method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes question generation and evaluation protocols.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: **No.**

   Justification: MIMIC-IV data is available under PhysioNet credentialed access (not open). Code will be released upon acceptance. Benchmark question templates and evaluation scripts will be included in the release.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: **Yes.**

   Justification: Section 4 specifies benchmark design, question counts, condition definitions, patient selection criteria, and evaluation methodology. Appendix N provides model versions and computational requirements.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly?

   Answer: **Yes.**

   Justification: Table 4 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Figure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample sizes ($n = 2$ patients per tier). The overall B2→B3 CI crossing zero is explicitly noted.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: **Yes.**

   Justification: Appendix N reports computational requirements: ClinicalBench ≈3.6 hours (single GPU for MedGemma inference), SliceBench ≈2 hours (Anthropic API).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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   Justification: The study uses de-identified clinical data under an approved data use agreement (Appendix T). No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous decision-making.

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    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: **Yes.**

    Justification: Section 6 discusses the potential for improved clinical decision support (positive) and risks of over-reliance on automated systems, context dilution, and the need for physician oversight (negative). Appendix T addresses data ethics.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

    Answer: **Yes.**

    Justification: The system operates on de-identified data under PhysioNet DUA. It is designed as a decision support tool requiring physician oversight. Code release will include usage guidelines emphasizing the system is not validated for autonomous clinical use.

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    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

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    Justification: MIMIC-IV is cited [31] and used under PhysioNet Credentialed Health Data Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open standard.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **Yes.**

Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Section 4, with question generation procedures, category definitions (Appendix O), and evaluation protocols. Templates and scoring scripts will accompany the code release.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

Answer: **N/A.**

Justification: No crowdsourcing was used. Physician adjudication (Appendix M) involves expert review by co-authors, not human subjects research.

15. **IRB approvals or equivalent**

Question: Does the paper describe potential risks and does it include the IRB approval number if applicable?

Answer: **N/A.**

Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not require separate IRB approval per PhysioNet policy.